

Creation Date 07-Apr-2009

Revision Date 25-Jan-2024

Revision Number 5

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                                  |                                               |
|----------------------------------|-----------------------------------------------|
| <b>Product Description:</b>      | <b>Boric acid</b>                             |
| <b>Cat No. :</b>                 | <b>36771</b>                                  |
| <b>Synonyms</b>                  | Boric acid; Orthoboric acid.; Hydrogen borate |
| <b>Index No</b>                  | 005-007-00-2                                  |
| <b>CAS No</b>                    | 10043-35-3                                    |
| <b>EC No</b>                     | 233-139-2                                     |
| <b>Molecular Formula</b>         | H3 B O3                                       |
| <b>REACH registration number</b> | -                                             |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                       |                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Recommended Use</b>                | Laboratory chemicals.                                                                       |
| <b>Sector of use</b>                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| <b>Product category</b>               | PC21 - Laboratory chemicals                                                                 |
| <b>Process categories</b>             | PROC15 - Use as a laboratory reagent                                                        |
| <b>Environmental release category</b> | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| <b>Uses advised against</b>           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

#### E-mail address

begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

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## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Based on available data, the classification criteria are not met

##### Health hazards

Reproductive Toxicity

Category 1B (H360FD)

##### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

### 2.2. Label elements



Signal Word

Danger

#### Hazard Statements

H360FD - May damage fertility. May damage the unborn child

#### Precautionary Statements

P201 - Obtain special instructions before use

P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

P308 + P313 - IF exposed or concerned: Get medical advice/attention

#### Additional EU labelling

Restricted to professional users

### 2.3. Other hazards

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component                                    | CAS No     | EC No     | Weight % | CLP Classification - Regulation (EC) No 1272/2008 |
|----------------------------------------------|------------|-----------|----------|---------------------------------------------------|
| Boric acid (H <sub>3</sub> BO <sub>3</sub> ) | 10043-35-3 | 233-139-2 | <=100    | Repr. 1B (H360FD)                                 |

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-

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                            |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if symptoms occur.                                                                                                                                                                                                                                |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                             |
| <b>Inhalation</b>                         | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                           |

### 4.2. Most important symptoms and effects, both acute and delayed

No information available.

### 4.3. Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive and/or toxic fumes.

#### **Hazardous Combustion Products**

Oxides of boron.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

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Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Do not get in eyes, on skin, or on clothing.

## 6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

## 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1D

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component          | European Union | The United Kingdom | France | Belgium                                                                 | Spain                                                                                           |
|--------------------|----------------|--------------------|--------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Boric acid (H3BO3) |                |                    |        | TWA: 2 mg/m <sup>3</sup> 8 uren<br>STEL: 6 mg/m <sup>3</sup> 15 minuten | STEL / VLA-EC: 6 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 2 mg/m <sup>3</sup> (8 horas) |

| Component          | Italy | Germany                       | Portugal                     | The Netherlands | Finland |
|--------------------|-------|-------------------------------|------------------------------|-----------------|---------|
| Boric acid (H3BO3) |       | TWA: 0.5 mg/m <sup>3</sup> (8 | STEL: 6 mg/m <sup>3</sup> 15 |                 |         |

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|  |  |                                                                                                                                                                                                                         |                                             |  |  |
|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--|--|
|  |  | Stunden). AGW - exposure factor 2<br>TWA: 10 mg/m <sup>3</sup> (8 Stunden). MAK when boric acid and tetraborates are present together, the MAK value is 0.75 mg boron/m <sup>3</sup><br>Höhepunkt: 10 mg/m <sup>3</sup> | minutos<br>TWA: 2 mg/m <sup>3</sup> 8 horas |  |  |
|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--|--|

| Component          | Austria | Denmark | Switzerland                                                                    | Poland | Norway |
|--------------------|---------|---------|--------------------------------------------------------------------------------|--------|--------|
| Boric acid (H3BO3) |         |         | STEL: 1.8 mg/m <sup>3</sup> 15 Minuten<br>TWA: 1.8 mg/m <sup>3</sup> 8 Stunden |        |        |

| Component          | Bulgaria                   | Croatia | Ireland                                                            | Cyprus | Czech Republic |
|--------------------|----------------------------|---------|--------------------------------------------------------------------|--------|----------------|
| Boric acid (H3BO3) | TWA: 5.0 mg/m <sup>3</sup> |         | TWA: 2 mg/m <sup>3</sup> 8 hr.<br>STEL: 6 mg/m <sup>3</sup> 15 min |        |                |

| Component          | Latvia                    | Lithuania                      | Luxembourg | Malta | Romania |
|--------------------|---------------------------|--------------------------------|------------|-------|---------|
| Boric acid (H3BO3) | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup> IPRD |            |       |         |

| Component          | Russia                    | Slovak Republic | Slovenia                                                                                                          | Sweden | Turkey |
|--------------------|---------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Boric acid (H3BO3) | MAC: 10 mg/m <sup>3</sup> |                 | TWA: 0.5 mg/m <sup>3</sup> 8 urah inhalable fraction<br>STEL: 1.0 mg/m <sup>3</sup> 15 minutah inhalable fraction |        |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                  | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|--------------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Boric acid (H3BO3)<br>10043-35-3 ( <=100 ) |                              |                                 |                                | DNEL = 392mg/kg bw/day            |

| Component                                  | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|--------------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Boric acid (H3BO3)<br>10043-35-3 ( <=100 ) |                                  |                                     |                                    | DNEL = 8.3mg/m <sup>3</sup>           |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component          | Fresh water    | Fresh water sediment | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture) |
|--------------------|----------------|----------------------|--------------------|------------------------------------|--------------------|
| Boric acid (H3BO3) | PNEC = 2.9mg/L |                      | PNEC = 13.7mg/L    | PNEC = 10mg/L                      | PNEC = 5.7mg/kg    |

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|                      |  |  |  |  |         |
|----------------------|--|--|--|--|---------|
| 10043-35-3 ( <=100 ) |  |  |  |  | soil dw |
|----------------------|--|--|--|--|---------|

| Component                                  | Marine water   | Marine water sediment | Marine water Intermittent | Food chain | Air |
|--------------------------------------------|----------------|-----------------------|---------------------------|------------|-----|
| Boric acid (H3BO3)<br>10043-35-3 ( <=100 ) | PNEC = 2.9mg/L |                       |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.  
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Natural rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                 |                          |       |
|---------------------------------|--------------------------|-------|
| <b>Physical State</b>           | Solid                    |       |
| <b>Appearance</b>               | White                    |       |
| <b>Odor</b>                     | Odorless                 |       |
| <b>Odor Threshold</b>           | No data available        |       |
| <b>Melting Point/Range</b>      | 169 °C / 336.2 °F        |       |
| <b>Softening Point</b>          | No data available        |       |
| <b>Boiling Point/Range</b>      | No information available |       |
| <b>Flammability (liquid)</b>    | Not applicable           | Solid |
| <b>Flammability (solid,gas)</b> | No information available |       |

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|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No information available | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | 100 °C                   |                                   |
| pH                                      | 3.8-4.8                  | 33 g/l aq.sol                     |
| Viscosity                               | Not applicable           | Solid                             |
| Water Solubility                        | Soluble                  |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Boric acid (H3BO3)                      | -0.757                   |                                   |
| Vapor Pressure                          | 2.7 mbar @ 20 °C         |                                   |
| Density / Specific Gravity              | No data available        |                                   |
| Bulk Density                            | No data available        |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Particle characteristics                | No data available        |                                   |

## 9.2. Other information

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | H3 B O3                |
| Molecular Weight  | 61.83                  |
| Evaporation Rate  | Not applicable - Solid |

## SECTION 10: STABILITY AND REACTIVITY

|                  |                                            |
|------------------|--------------------------------------------|
| 10.1. Reactivity | None known, based on information available |
|------------------|--------------------------------------------|

|                          |                     |
|--------------------------|---------------------|
| 10.2. Chemical stability | Moisture sensitive. |
|--------------------------|---------------------|

## 10.3. Possibility of hazardous reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

|                           |                                                                                 |
|---------------------------|---------------------------------------------------------------------------------|
| 10.4. Conditions to avoid | Incompatible products. Excess heat. Avoid dust formation. Exposure to moisture. |
|---------------------------|---------------------------------------------------------------------------------|

|                              |                                        |
|------------------------------|----------------------------------------|
| 10.5. Incompatible materials | Strong oxidizing agents. Strong bases. |
|------------------------------|----------------------------------------|

|                                        |                  |
|----------------------------------------|------------------|
| 10.6. Hazardous decomposition products | Oxides of boron. |
|----------------------------------------|------------------|

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| (a) acute toxicity; |                                                                  |
| Oral                | Based on available data, the classification criteria are not met |
| Dermal              | Based on available data, the classification criteria are not met |
| Inhalation          | Based on available data, the classification criteria are not met |

| Component          | LD50 Oral          | LD50 Dermal             | LC50 Inhalation |
|--------------------|--------------------|-------------------------|-----------------|
| Boric acid (H3BO3) | 2660 mg/kg ( Rat ) | > 2000 mg/kg ( Rabbit ) | Not listed      |

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|                                                                                                   |                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (b) skin corrosion/irritation;                                                                    | Based on available data, the classification criteria are not met                                                                                                                                                                       |
| (c) serious eye damage/irritation;                                                                | Based on available data, the classification criteria are not met                                                                                                                                                                       |
| (d) respiratory or skin sensitization;<br>Respiratory<br>Skin                                     | Based on available data, the classification criteria are not met<br>Based on available data, the classification criteria are not met                                                                                                   |
| (e) germ cell mutagenicity;                                                                       | Based on available data, the classification criteria are not met                                                                                                                                                                       |
| (f) carcinogenicity;                                                                              | Based on available data, the classification criteria are not met<br>There are no known carcinogenic chemicals in this product                                                                                                          |
| (g) reproductive toxicity;<br>Reproductive Effects<br>Developmental Effects<br><br>Teratogenicity | Category 1B<br>Adverse reproductive effects have occurred in humans.<br>May cause harm to the unborn child. Developmental effects have occurred in experimental animals.<br>Teratogenic effects have occurred in experimental animals. |
| (h) STOT-single exposure;                                                                         | Based on available data, the classification criteria are not met                                                                                                                                                                       |
| (i) STOT-repeated exposure;<br><br>Target Organs                                                  | Based on available data, the classification criteria are not met<br><br>None known.                                                                                                                                                    |
| (j) aspiration hazard;                                                                            | Not applicable<br>Solid                                                                                                                                                                                                                |
| Symptoms / effects,both acute and delayed                                                         | No information available.                                                                                                                                                                                                              |

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

**Ecotoxicity effects** Do not empty into drains. .

| Component          | Freshwater Fish                       | Water Flea                                | Freshwater Algae |
|--------------------|---------------------------------------|-------------------------------------------|------------------|
| Boric acid (H3BO3) | Gambusia affinis: LC50: 5600 mg/L/96h | EC50: 115 - 153 mg/L, 48h (Daphnia magna) | -                |

| Component          | Microtox | M-Factor |
|--------------------|----------|----------|
| Boric acid (H3BO3) | -        |          |

### 12.2. Persistence and degradability

**Persistence** Persistence is unlikely.  
**Degradability** Not relevant for inorganic substances.

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## 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component                                    | log Pow | Bioconcentration factor (BCF) |
|----------------------------------------------|---------|-------------------------------|
| Boric acid (H <sub>3</sub> BO <sub>3</sub> ) | -0.757  | 0 dimensionless               |

## 12.4. Mobility in soil

The product is water soluble, and may spread in water systems . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

## 12.5. Results of PBT and vPvB assessment

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

#### Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### ADR

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### IATA

Not regulated

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## 14.1. UN number

## 14.2. UN proper shipping name

## 14.3. Transport hazard class(es)

## 14.4. Packing group

## 14.5. Environmental hazards

No hazards identified

## 14.6. Special precautions for user

No special precautions required.

## 14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component          | CAS No     | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------------|------------|-----------|--------|-----|-------|------|----------|------|------|
| Boric acid (H3BO3) | 10043-35-3 | 233-139-2 | -      | -   | X     | X    | KE-03499 | X    | X    |

| Component          | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|--------------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Boric acid (H3BO3) | 10043-35-3 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

| Component          | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------|------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Boric acid (H3BO3) | 10043-35-3 | -                                                                   | Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - 233-139-2 - Toxic for reproduction, Article 57c                                 |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

#### REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

#### Seveso III Directive (2012/18/EC)

| Component          | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Boric acid (H3BO3) | 10043-35-3 | Not applicable                                                                            | Not applicable                                                                           |

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

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Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component          | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|--------------------|---------------------------------------|-------------------------|
| Boric acid (H3BO3) | WGK1                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H360FD - May damage fertility. May damage the unborn child

H360Fd - May damage fertility. Suspected of damaging the unborn child

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

# SAFETY DATA SHEET

Boric acid

Revision Date 25-Jan-2024

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

|                  |                                                    |
|------------------|----------------------------------------------------|
| Prepared By      | Health, Safety and Environmental Department        |
| Creation Date    | 07-Apr-2009                                        |
| Revision Date    | 25-Jan-2024                                        |
| Revision Summary | New emergency telephone response service provider. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

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**End of Safety Data Sheet**